VibeCheck

An Explainable AI Framework for Cross-Platform Interest Modeling and Personalized Recommendations

Abstract

With the increasing use of digital platforms such as Spotify, YouTube, and Instagram, users generate large amounts of digital activity every day. However, these platforms operate independently and only provide platform-specific recommendations and summaries, preventing users from understanding their overall interests and digital behavior.

This project proposes **VibeCheck**, an Explainable AI-based cross-platform recommendation framework that combines user activities from multiple platforms to create a unified interest profile. Using semantic language models, the system identifies user interests and generates personalized cross-platform recommendations, such as recommending podcasts, books, creators, or communities based on interests identified from another platform. Unlike traditional recommendation systems, VibeCheck explains why each recommendation is made, increasing user trust and transparency.

In addition to recommendations, the system generates weekly or monthly AI-powered recap reports, enabling users to reflect on their evolving interests and digital habits. Privacy is maintained through anonymous community participation and private friend groups. The proposed framework aims to provide meaningful recommendations while promoting digital reflection through explainable artificial intelligence.

1. Introduction

Digital platforms have transformed the way people learn, communicate, and consume content. Applications such as Spotify, YouTube, and Instagram continuously collect valuable information about users' interests and online behavior. However, each platform works independently and only provides recommendations based on activities within its own ecosystem.

As a result, users receive fragmented recommendations and isolated summaries instead of a complete understanding of their interests. Existing recap systems mainly display usage

statistics, while recommendation systems focus primarily on increasing user engagement rather than providing meaningful, explainable suggestions across platforms.

To address this limitation, this project proposes **VibeCheck**, an Explainable AI framework that combines user activities from multiple platforms into a unified interest profile. Based on this profile, the system generates cross-platform recommendations with clear explanations, helping users discover relevant content, creators, books, and communities beyond a single platform. Additionally, AI-generated recap reports allow users to understand how their interests evolve over time, promoting digital reflection and better content discovery.

2. Problem Statement

Current digital platforms operate independently, resulting in isolated recommendation systems that do not consider a user's complete digital interests. Existing recap applications mainly summarize statistics without providing meaningful insights or personalized recommendations. Furthermore, most recommendation systems lack transparency, making it difficult for users to understand why certain content is suggested.

There is a need for a unified framework capable of integrating user activities across multiple platforms, generating explainable cross-platform recommendations, and providing personalized reflections while maintaining user privacy.

3. Objectives

The proposed system aims to:

- Collect user activity from multiple digital platforms.
- Build a unified cross-platform interest profile.
- Generate personalized cross-platform recommendations.
- Provide natural language explanations for every recommendation.
- Generate weekly or monthly AI-powered recap reports.
- Recommend relevant creators, books, content, and communities.
- Enable anonymous community participation and private friend groups.
- Promote digital reflection while maintaining user privacy.

4. Scope of the Project

The proposed system focuses on analyzing user interests across multiple digital platforms to generate personalized recommendations.

The scope includes:

- Spotify music and podcast preferences.
- YouTube watch history and subscriptions.
- Instagram interests through exported activity.
- AI-based interest extraction.
- Cross-platform recommendation generation.
- Explainable recommendation system.
- Weekly and monthly recap generation.
- Anonymous communities.
- Private friend groups.
- Shareable recap cards.

Future versions can include Reddit, Netflix, Goodreads, GitHub, and other digital platforms.

5. Literature Survey

Several researchers have proposed systems for digital wellbeing, recommendation systems, explainable AI, and privacy-preserving recommendations. However, these approaches generally focus on only one aspect of the problem.

- **Understanding Digital Wellbeing Through Smartphone Usage Intentions (IEEE, 2024)** analyzed smartphone behavior to understand digital wellbeing but did not provide cross-platform recommendations.
- **Recommendation with Dynamic Natural Language Explanations (IEEE, 2023)** generated explainable recommendations but focused only on improving recommendation transparency rather than cross-platform interest analysis.
- **KEMB-Rec (IEEE, 2024)** modeled user behavior for explainable recommendations but did not include digital reflection or recap generation.
- **Privacy Protection in Cross-Platform Recommender Systems (Springer, 2023)** discussed privacy-preserving recommendation techniques but did not integrate AI reflections or explainable recommendations.

Research Gap

Most existing systems focus on recommendation, digital wellbeing, or privacy independently. Very few integrate these components into a single framework. Additionally, current systems generally provide recommendations within the same platform rather than across different platforms.

VibeCheck addresses this gap by combining cross-platform interest modeling, explainable AI, AI-generated reflections, and privacy-preserving social interaction within a unified framework.

6. Proposed Methodology

The proposed framework follows a multi-stage pipeline.

Step 1 – Data Collection

The system collects user activity from Spotify, YouTube, and Instagram through APIs or exported user data.

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Step 2 – Interest Extraction

The collected activities are analyzed to identify meaningful user interests such as AI, music, fitness, photography, programming, or travel.

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Step 3 – Pre-trained Language Model

A Sentence Transformer converts extracted interests into semantic embeddings, enabling the system to understand relationships between similar topics without requiring a custom-trained model.

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Step 4 – Unified User Interest Profile

Interests from all platforms are combined into a single user profile, representing the user's overall preferences instead of platform-specific behavior.

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Step 5 – Cross-Platform Recommendation Engine

Using the unified profile, the system recommends:

- **Content:** Videos, music, podcasts.
- **Creators:** YouTubers, artists, podcasters.
- **Books:** Based on user interests.
- **Communities:** Relevant discussion groups and forums.

Recommendations are generated across platforms rather than within a single application.

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Step 6 – Explainable AI Module

Every recommendation is accompanied by a natural language explanation.

Example:

"This podcast is recommended because you frequently watch Artificial Intelligence tutorials on YouTube."

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Step 7 – Weekly Reflection & Recap

The system generates weekly or monthly recap reports highlighting:

- Top interests.
- Interest evolution.
- Cross-platform activity.
- Personalized AI reflections.
- Recommended content.

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Step 8 – Social Features

Users can:

- Share recap cards.
- Join anonymous interest-based communities.
- Create private groups with friends.

7. System Flow

Spotify / YouTube / Instagram
↓
Data Collection
↓
Interest Extraction
↓
Pre-trained Language Model
(Sentence Transformer)
↓
Interest Embeddings
↓
Unified User Interest Profile
↓
Cross-Platform Recommendation
↓
Explainable AI Generator
↓
Digital Reflection & Wellbeing Analysis
↓
Weekly Reflection & Recap
↓
Communities & Private Groups

8. Tools & Technologies

Component	**Technology**
Frontend	React.js
Backend	FastAPI
Database	MongoDB
AI	Sentence Transformers
Recommendation	Cosine Similarity, Top-K Ranking
Explainability	LLM (e.g., GPT/Llama)
Visualization	Plotly / Chart.js

APIs	Spotify API, Google Takeout, Instagram Export

9. Expected Outcome

The proposed system will generate personalized cross-platform recommendations by combining user interests from multiple platforms. Users will receive transparent recommendations with AI-generated explanations instead of platform-specific suggestions. The system will also encourage digital reflection through weekly recap reports while preserving privacy using anonymous communities and private groups.

10. Conclusion

VibeCheck proposes an Explainable AI framework for generating cross-platform recommendations through unified user interest modeling. Unlike traditional recommendation systems that operate independently within individual platforms, the proposed system integrates multiple digital platforms to provide meaningful, explainable, and personalized recommendations. By combining recommendation, AI-generated reflections, and privacy-preserving social features, VibeCheck offers a user-centered approach to content discovery and digital awareness.

11. References (Sample)

1. *Understanding Digital Wellbeing Through Smartphone Usage Intentions and Regrettable Patterns*, IEEE, 2024.
2. *Recommendation with Dynamic Natural Language Explanations*, IEEE, 2023.
3. *KEMB-Rec: Knowledge-Enhanced Explainable Multibehavior Recommendation*, IEEE, 2024.
4. *Privacy Protection in Cross-Platform Recommender Systems*, Springer, 2023.
5. *Explainable Recommender Systems: From Theory to Practice*, Springer, 2023.